1 **(a)** $\left(\frac{-2+4}{2}, \frac{5+(-3)}{2}\right)$ **[M1]**

$(1, 1)$ **[A1]**

(b) $(4-(-2))^2 + (-3-5)^2$ or $6^2 + (-8)^2$ **[M1]**

$\sqrt{36+64}$ or $\sqrt{100}$ **[M1]**

10 **[A1]**

[Total Marks: 5]

2 **(a)** $\left(\frac{1+7}{2}, \frac{4+(-4)}{2}\right)$ **[M1]**

$(4, 0)$ **[A1]**

(b) $(7-1)^2 + (-4-4)^2$ oe **[M1]**

$\sqrt{36+64}$ or $\sqrt{100}$ **[M1]**

10 **[A1]**

[Total Marks: 5]

3 **(a)** Calculates length OP: $\sqrt{(-3)^2+4^2}$ or $\sqrt{9+16}$ **[M1]**

5 **[A1]**

(b) Recognises $O(0, 0)$ is the midpoint of PQ, so sets up $\frac{-3+x}{2} = 0$ and $\frac{4+y}{2} = 0$ OR uses translation from P to O **[M1]**

$(3, -4)$ **[A1]**

(c) Sets up equation for length OR: $k^2 + (\sqrt{11})^2 = 5^2$ **[M1]**

$k^2 + 11 = 25$

$k^2 = 14$ [M1]

$-\sqrt{14}$ **[A1]**

[Total Marks: 7]

4 **(a)** $\left(\frac{-1+5}{2}, \frac{2+8}{2}\right)$ **[M1]**

$(2, 5)$ **[A1]**

(b) $(2-(-1))^2 + (5-2)^2$ or $\frac{1}{2}\sqrt{(5-(-1))^2 + (8-2)^2}$ oe **[M1]**

$\sqrt{9+9}$ or $\sqrt{18}$ **[M1]**

$3\sqrt{2}$ **[A1]**

(c) $\frac{2+x}{2} = \frac{5}{2}$ and $\frac{5+y}{2} = \frac{3}{2}$ oe **[M1]**

$x = 3$ and $y = -2$ to give $D(3, -2)$ **[A1]**

Length $AD = \sqrt{(3-(-1))^2 + (-2-2)^2}$ oe **[M1]**

$\sqrt{16+16}$ or $\sqrt{32}$ **[M1]**

$4\sqrt{2}$ **[A1]**

[Total Marks: 10]

5 **(a)** Uses distance formula: $\sqrt{(4-(-2))^2+(-3-5)^2}$ **[M1]**

Simplifies to $\sqrt{6^2+(-8)^2}$ or $\sqrt{36+64}$ **[M1]**

10 **[A1]**

(b) Uses midpoint formula: $\left(\frac{-2+4}{2}, \frac{5+(-3)}{2}\right)$ **[M1]**

$(1, 1)$ **[A1]**

[Total Marks: 5]

6 **(a)** $(7-3)^2+(5-k)^2=5^2$ or equivalent **[M1]**

$16+(5-k)^2=25$ or $(5-k)^2=9$ **[M1]**

$k=2$ and $k=8$ **[A1]**

(b) Identifies $P(3,2)$ to use in midpoint calculation: $\left(\frac{3+7}{2}, \frac{2+5}{2}\right)$ **[M1]**

$(5, 3.5)$ **[A1]**

[Total Marks: 5]

7 **(a)** $(k-(-3))^2+(-1-5)^2=10^2$ oe **[M1]**

$(k+3)^2+36=100$ or $k^2+6k+9+36=100$ **[M1]**

$(k+3)^2=64$ or $k^2+6k-55=0$ **[M1]**

$k=5$ or $k=-11$ **[A1]**

(b) Chooses $k=5$ to establish coordinates of Q as $(5,-1)$ **[M1]**

$(1, 2)$ **[A1]**

[Total Marks: 6]

8 **(a)** $\left(\frac{-3+5}{2}, \frac{2+8}{2}\right)$ **[M1]**

$(1, 5)$ **[A1]**

(b) Radius $r=\sqrt{(1-(-3))^2+(5-2)^2}$ or $\sqrt{4^2+3^2}=5$ **[M1]**

Area $=\pi\times5^2$ **[M1]**

78.5 **[A1]**

(c) Distance $CD=5 \implies (k-1)^2+(8-5)^2=5^2$ **[M1]**

$(k-1)^2+9=25 \implies (k-1)^2=16$ **[M1]**

$k = 5$ and $k = -3$ **[A1]**

[Total Marks: 8]

9 **(a)** Sets up the distance formula: $\sqrt{(3p-2p)^2+(4p-p)^2} = 5\sqrt{10}$ **[M1]**

Simplifies the terms inside the root: $\sqrt{p^2+(3p)^2} = 5\sqrt{10}$ **[M1]**

Squares both sides and simplifies: $10p^2 = 250$ **[M1]**

$p = 5$ (rejects -5 as $p > 0$) **[A1]**

(b) Evaluates coordinates of $A(10, 5)$ and $B(15, 20)$ **[B1]**

Sets up a valid expression for the coordinates of C, e.g., x-coordinate $= 10 + \frac{2}{5}(15-10)$ or y-coordinate $= 5 + \frac{2}{5}(20-5)$ **[M1]**

$(12, 11)$ **[A1]**

(c) Uses the midpoint formula, e.g. $\frac{15+x}{2} = 12$ and $\frac{20+y}{2} = 11$, or $D = 2C - B$ **[M1]**

$(9, 2)$ **[A1]**

[Total Marks: 9]